



by Hana Alamuddin

Selangor Turf Club

Kuala Lumpur, Malaysia



Architects

*T.R. Hamzah & Yeang
Lobb Partnership*

Client

Selangor Turf Club

Design

1989-1991

Completed

1993

I. Introduction

The Selangor Turf Club is one of the most modern and extensively equipped racetracks in Asia. The site is located south of Kuala Lumpur and north of Serdang, in the centre of a recreation zone, and is spread over 103 hectares (255 acres) of infill land that was previously an abandoned tin mine. The facilities include: racing tracks, a 25'000 capacity grandstand, stables for up to 700 horses, an equine pool and equine clinic, as well as an equestrian sports club with various training tracks.

II. Contextual Information

a. Historical background

The Selangor Turf Club (STC) was established in 1896 by sporting enthusiasts and horse lovers who introduced thoroughbred racing to Malaysia. The club is private and has about 500 members including local businessmen, horse owners, and many of the community's social elite.

In the 1980s the Prime Minister decided to relocate the club out of the city centre in order to develop the site, now the location of Cesar Pelli's twin Petronas towers, and to help develop a derelict area.

Four local architects in partnership with four experts on racetracks were asked for fee bids and interviewed. The winners were Ken Yeang in partnership with Lobb Partners.

The old site was sold and the new site was bought at MYR 18'375'547.

In the new location the clients wanted to introduce a multi-purpose facility to promote a "new era for the local equine industry."

b. Local architectural character

In general, the local architecture in and around Kuala Lumpur is quite eclectic, a mix of British colonial style and Chinese influences, especially the "shop house." The post-war period was characterised by European influence in architecture and following this era came the Malaysian period, where many high-rise buildings with high plot ratios were introduced. A recent and significant style is the high-tech building and public buildings are designed in what is locally known as the "Moorish" style while sports buildings tend to be simple modern structures with large roofs, quite utilitarian in character.

c. Climatic conditions

Malaysia lies between latitudes 1° and 7° North, and longitudes 100° and 119° East, and is therefore in the tropics. Its climate is both hot and humid with heavy rainfall during April/ May and October/ November. Air temperatures are uniformly high throughout the year with a mean maximum temperature of 29°C to 32°C during the day and a mean minimum of 22°C to 24°C at night in the coastal areas. The prevailing winds tend to come from the northeast but due to the nature of the peninsula change according to the location. The average annual rainfall is between 250 and 300 cm.

d. Site context

The Selangor Turf Club is in the Sungai Besi's recreation zone. This region is also home to the Bukit Jalil site which contains the Commonwealth village and stadium for the 1998 games. There is also the Mine Resort which has been transformed into a golf course and theme park. The closest development to the turf club however, is the recently built Golden Horses hotel which is built in the Moorish style.

e. *Site topography*

The site is very open and was previously an abandoned tin mine. It required extensive cut and fill, draining and levelling, and filling the pond that existed in its middle. That alone took six months to achieve. The grandstand, however, is built on solid ground.

III. Programme

a. *Conditions of programme formulation*

The relocation of the club led to an expansion of its programme from just a racing facility to a centre for the promotion of equestrian activities and a more family-oriented place. This was achieved through the addition of an equestrian and sports club which offers riding lessons for a small fee, a multi-purpose hall, two auditoriums, a country club (unbuilt), and other sports facilities.

b. *Objectives*

The main objective was to design a first-class racing facility that would also allow flexibility in use and possible expansion to meet the client's future needs. An efficient security/circulation system was crucial to the workings of the project as well as an energy efficient design that would reduce the operational cost of the scheme.

c. *Functional requirements*

The architect and the racing consultant formulated the brief document. The client wanted complete separation between the members of the club and the public. Each had to have their own parking, entrance lobbies, betting areas, and facilities. The client required twelve corporate boxes for rent and a prestigious members entrance. The security/circulation of the project required that the circulation of money does not cross the circulation of people, the circulation of services not cross either, and the circulation of the horses to be completely separate to protect the horses from the risk of getting drugged.

IV. Description

a. *Project data*

The grandstand, with a total seating capacity of 25'000 people, is divided into two parts, the members' section and the public section. The members have their own entrance lobby with an escalator that goes to the top, private section of the building. There they have their own facilities and watch the races in their air-conditioned space (capacity 1'200) or from the twelve executive boxes. On either side of the member's entrance are restaurants that cater to the members but are also open to the public on non-racing days.

The public approach the building from the covered walkway in the parking lot to the ticketing hall. From the ticket area they continue under the shaded area to the grandstand. They enter the building at the level of the hawkers or fast-food hall. They either go down one level to the tote or multi-purpose hall or up to the grandstand. A section of the public grandstand is also air-conditioned and can be entered by buying a special ticket. The tote hall has betting windows to one side and a 24-monitor video wall system and extensive television monitors. During non-race days this hall is rented out for trade fairs and amateur sports activities.

Under the grandstand are also the administrative offices, two auditoriums, the jockey's centre, service areas, and an extensive audio-visual facility that allows the broadcasting of races straight onto national television. The facility also allows for the close monitoring of jockeys, the photo finish, and security around the project.

The Track

The track was required to have a 1'000-m training straight. It has a running area of 2'000 m on the outer bend and 1'800 m on the inner bend. It has a paper-clip shape with extended ends to meet the length requirements within the site. It has both grass and sand training tracks and because of its elevated bends it successfully reduces the leg and tendon injuries of the horses.

Opposite the grandstand is the presentation dais. Inside the track is the infield area which is lower than the track so as not to obstruct the back-straight run from view. Various sports facilities are being planned for this area. Today a cricket pitch, donated by the club, is being prepared for the 1998 Commonwealth Games. The water feature is also used for irrigation.

Other racing facilities include the visitors stables, the residents stables, the parade ring, the saddling stables, and other racing and training areas including the veterinarian's clinic and the equine pool. The stables are rented out to the trainers for a very small fee as a service to the industry.

The STC Equestrian and Sports Centre was officially opened on 4th April 1994. The new direction of the club caters to equestrian and sporting enthusiast of all ages. It is located in the visitors' stables section of the project and quite separate from the other facilities. For a very small fee, the centre offers the public lessons in dressage, show jumping, and even a simulator horse. The club has become the official venue for local equestrian championships and international horse shows. However, the highlight of the equestrian calendar is the National Horse Show. During this event various carnival activities, catering to the entire family, take place to introduce the public to the world of the horse.

Infrastructure

The whole complex is surrounded by an access road that leads to the various parts of the project. The members' and trainers' car parking lots are south of the grandstand and have a total capacity of 500 cars. The public car park to the north of the grandstand holds 4'500 cars and 2'000 motorcycles. Located in the northwest area of the site are the electrical substations, sewage treatment plant, and accommodations for the various workers.

Grandstand building data

This is a five-storey concrete structure:

Gross Floor Area

Level 1:	4'055 m ²
Level 2:	9'194 m ²
Level 3:	1'688 m ²
Level 4:	4'131 m ²
Level 5:	2'449 m ²
Total	34'462 m ²

Project components:	1 grandstand capacity 25'000 people
	700 stables (480 local and 220 visitors)
	27 trainers' offices
	3 lunging boxes
	5 rolling boxes
	20 saddling stalls
	20 open paddock shelters
	1 starting stall shelter
	5 camera towers
	1 ticketing hall, police holding cells, and security centre
	1 guard house
	9 consumer sub-stations
	1 covered walkway
	1 veterinarian's clinic
	1 equine pool
	2 auditoriums (2 x 90)
	4'500 public car park
	450 member's car park
	2'000 motorcycle parking area

b. *Evolution of design concepts*

The response to the setting is strictly functional. In order to avoid the glare of the afternoon sun, the track is oriented along the north-south axis. The grandstand faces East for the same reason and is positioned at a five-degree angle to the track to ensure the best sight lines for the start and finish. The rest of the facilities simply fall into place according to the circulation of the horses. The visitors' stables and riding school are at the south end of the site. The residents' stables and the various training facilities such as the open paddock are at the north end. The veterinarian's clinic and equine pool are to the East of the track.

For the races, the horses come from their stables to the saddling stalls south of the grandstand. They are saddled and then paraded and mounted in front of the punters in the parade ring, after which they proceed to the start line. At the end of the race, the first four horses come back into winner's enclosure inside the parade ring and the others go back to the saddling stalls where the jockeys dismount and go to their special area under the grandstand. The horses return to their stables.

Climatically the building is designed to make use of the Venturi effect for natural ventilation. The effect is based on the idea that when an enclosure is provided with openings at the top and bottom, the hot air rises and is released at the top creating a vacuum that pulls cold air in from the lower openings thereby creating air movement. Hence the roof of the grandstand is designed as a series of leaves with open louvers between them to allow the air to escape without letting the rain in. The overhang of the roof also shades the glass from the sun and heat gain. The grandstand building, however, is not climatically designed, according to the architect, it is simply naturally ventilated. Thirty percent of the area of the grandstand is air-conditioned. Even the stables are naturally ventilated with roof ventilators.

The clients' objectives of flexibility and expansion were achieved through the use of a standard structural section in the seating area of the grandstand that would allow the expansion or contraction of the air-conditioned areas. The linear "spine" of services and circulation easily allows for the future extension of the building. Service areas are designed, it seems, to cater for future expansion too. Plans are such that the grandstand would extend North while a country club would be built to the South. Within the building the location of the services on the edges of the large areas also plays into the flexible use of the building and its future expansion.

The issue of security and circulation operates simultaneously at many levels. This was resolved by careful planning and zoning. "The zoning of the building, both horizontally and vertically, provides visitors with a hierarchy of security from the King to the average punter." T. R. Hamzah & Yeang

The building envelope is very simple with a minimum of obstructed views. The roof is the main feature as it sails over the grandstand on its steel trusses. The only articulated detail in the building is the steel connection to the structure.

The landscaping of the project is formal and restricted only to certain areas and simply plays a decorative role adding colour to the track and grandstand.

Structure, Materials, Technology

The site earthworks were a major part of the works and cost of the project. The cut and fill and draining of the land was extensive.

Due to the nature of the site, piling was used for the foundations. The superstructure of the buildings is concrete frame with concrete masonry unit (CMU) infill. The frame of the grandstand is made of prestressed and post-tensioned concrete beams and slabs. The roofs are steel trusses with steel roofing panels, which were assembled on site and lifted into position. The cantilevered, steel roof of the grandstand created a "spectator-friendly" column-free space. Steel was chosen due to a tight construction schedule and the lack of temperature differential in Malaysia makes the detailing of steel fairly easy as expansion and movement joints are not necessary. The steel trusses are sand blasted and painted. Steel construction was still fairly new in Malaysia when this project was built. The glass is tinted because of the limited budget, the double glazing requested by the architect was not installed.

The mechanical and electrical system in the grandstand building is a simple fan coil system that allows maximum flexibility. The building also has the latest Building Automation System (BAS) and a computer that controls energy consumption. Also the extensive network of closed caption television (CCTV) was mentioned previously.

Origin of technology, materials, labour force, professionals

Steel technology had only recently been introduced in Malaysia at the time of the project. The architect himself learned about steel detailing from the structural engineer which he then used in various other projects such as the Menara Mesiniaga building. The materials were mainly imported – the glass and steel pipes from Japan, the marble from Italy.

The labour force was comprised of 60 percent skilled workers and 40 percent unskilled workers. The origin of the labour force was 30 percent domestic and 70 percent foreign (mainly Indonesian).

The architects were a local firm in alliance with a British racing consultant. Other consultants were local and Australian and the contractors were Japanese.

Construction Schedule and Costs

History of project

After being selected, the Lobb Partnership and T. R. Hamzah & Yeang became partners for the project. The Selangor Turf Club raised 80 percent of the project budget from the sale of the old club property and commissioned these associates in 1989. Project design commenced in 1989 and

construction began in 1991. Completion and occupancy of the project took place in 1993.

b. Total costs and main sources of financing

The sources of financing were internal, mainly from the sale of the old club property.

Land cost:	MYR 18'315'547
Total construction cost:	MYR 48'000'000

c. Comparative costs

For comparison, the cost per square metre for Shah Alam Stadium (only the building, excluding the athletic track and field) is MYR 1'500 according to the project's quantity surveyor.

d. Qualitative analysis of costs

Contract Cost	Final Cost (MYR)
Earthworks	31'514'494
Piling	1'818'810
Grandstand	56'604'601
Substation	835'659
Infrastructure	31'091'304
Tracks and infield	24'309'360
Soft landscaping	1'311'321
Specialists equipment	1'328'407
Stables	21'242'131
Ancillary building	10'558'243
Sewage treatment plant	772'154.00
Electronic security system	2'808'671
LPG reticulation	141'122
Site supervision	1'542'454
Consultants' fees (forecast)	22'558'280
Land cost	18'375'547
Total	230'968'299

Cost per square metre: USD 617

e. Maintenance costs

Air conditioning and ventilation	3'333'559
Housekeeping/maintenance	52'000 per month
M & E maintenance	45'865 per month

VI. Technical Assessment

a. Functional assessment

The building is functionally very successful. The jockeys find the track very well designed and say it is the best around. The club administrators find the project very easy to run and efficient because the administrative section of the club is in one building, under the grandstand. The separation between the members and the public is achieved without denying the members the special experience and

excitement of the races as their circulation area overlooks the food hall and the public. During a race this activity gives the whole building a sense of the intensity of the sport.

The layout of the master plan makes operation during the races very smooth and easy. However, there seemed to be a lot of punters sitting on the floor and the public spaces got quite dirty. This however did not seem to dampen the spirit of the punters.

b. Climatic performance

Generally, air movement and ventilation is good even on race days. Twelve fans were added to cope with the large number of heavy smokers. Sometimes, during a period of heavy rain, water gets into the interior of the building.

c. Choice of materials, level of technology

The architect's decision to give a high-tech image to his building and the tight construction schedule, governed the choice of materials. The detailing is neat and articulates the structure when required. The main elements of the steel trusses came in several sections in length but these were welded to look like one piece of steel pipe. The problem with the steel work seems to have been with the steel sheeting used as a roof cover. Provided by a Japanese supplier, these sheets rusted badly. Whether the reason was acid rain, bad workmanship, or the use of a non-standard colour, was never determined. Finally, a compromise was reached between the client and the supplier and the sheets were repainted. The steel structure seems fine in the grandstand building but rusting was noticed in the resident's stables. Galvanisation would probably have been a much better finish for steel in this climate.

d. Ageing and maintenance

It seems that no major maintenance or refurbishment has been done on the building since its completion (apart from the steel sheets). The grandstand is in good condition overall and work is proceeding on upgrading the finishes in the facility now that new funds are available. Some external walls require repainting but apart from the steel in the stables the facilities appear to be in good shape. The project however did suffer from some settlement which seems to be ending. In the last six months the land settled one inch as opposed to six and eight inches in the previous six-month periods. There is worse settling in the public car park where even trees are not growing successfully.

Another problem that appeared was with two of the camera towers which vibrated and affected the cameras. Steel tension wires were added to stabilise the structures.

The only maintenance problem is with the high ceiling of the grandstand. In certain parts, it is too high for even scaffolding or a scissors lift. Moreover these areas of the building seem to be popular with the birds!

e. Design features

The massing and integration of the project into the site is very successful. The break-up and the form of the roof help the volume of the grandstand to sit well on the site. The variety of volumes and different spaces inside the building create a sense of spatial play and complexity that give the building its architectural quality.

The landscaping, however, leaves a lot to be desired. Trees and planting could have considerably improved the environment. Even near the stables, trees are scarce and function more as decoration rather than to provide shade for the horses while they move around the site.

VII. Users

The members are mainly from the social elite or upper classes. The public is mostly working-class Chinese. As previously mentioned the clients are very happy with the facility. The public seem to be well catered to and it is becoming very popular with the tourists, especially those from Taiwan. The restaurants operate seven days a week and seem quite popular. The riding facility is also popular with the community.

VIII. Persons involved

Project personnel

- WTW Consultant BHD, Project Manager
- Lobb Partnership, Racing Consultant
- Tahir Wong SDN BHD, Structural Engineers
- Engineering & Environmental, Civil Engineers
- Consultants SDN BHD ER/Nair Associates/Norman Disney & Young SDN BHD, Mechanical & Electrical Engineers
- Konsultant Perkid Matan Kontrek, Quantity Surveyor
- T.R. Hamzah & Yeang Design SDN BHD, Interior Designer & Architects
- Malik Lip Associates, Landscape Consultant
- AV Technology, Acoustics Consultant
- Sato Kogyo, Main Builder

Hana Alamuddin

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